BECOMING A DOCTOR OF

INTERNATIONAL ENERGY AND NATURAL RESOURCES LAW

DOCTORAL TEXT AND ADVANCED PROFESSIONAL REFERENCE



TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

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International Energy and Natural Resources Law

A Doctoral Text and Advanced Professional Reference on Resource Sovereignty, Energy Transactions, Investment Protection, Environmental and Climate Governance, and International Dispute Resolution

TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

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Becoming a Doctor of International Energy and Natural Resources Law

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DEDICATION

To the scholars, regulators, negotiators, advocates, and arbitrators whose rigorous work turns the world's energy and natural resources into lawful, accountable, and equitable development —

and to the communities whose lands, waters, and futures that work must always serve.

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PART I

Jurisprudential and Institutional Foundations

The conceptual architecture of the discipline: its definition and history, its sources and principles, the doctrine of permanent sovereignty over natural resources, and the institutions through which global energy governance is organised.

1 CHAPTER

The Nature, Scope, and Evolution of International Energy and Natural Resources Law

LEARNING OBJECTIVES

After studying this chapter, the reader should be able to:

1. Define international energy and natural resources law as a field of legal study and distinguish it from adjacent fields, including public international law, international economic law, and international environmental law.
2. Trace the historical development of resource governance from the early concession era through decolonisation, the New International Economic Order, liberalisation, and the contemporary energy transition.
3. Explain how the internationalisation of energy markets has transformed the regulation of exploration, production, transportation, and trade.
4. Analyse the interaction between public law, private law, and transnational legal relations in energy and resource projects.
5. Evaluate the significance of legal pluralism and competing regulatory authorities in global resource governance.
6. Identify the principal foundational debates that structure doctoral research in the field.

Introduction

Energy and natural resources sit at the junction of almost every question that modern legal systems are asked to answer. Who owns what lies beneath the soil and the seabed? Under what conditions may a sovereign state grant rights to explore for and extract those resources, and on what terms may it later change its mind? How should the economic rent produced by a non-renewable asset be shared between the state, the investor, the host community, and future generations? What obligations attach to the transport of oil and gas across national frontiers, to the generation and trade of electricity, or to the emissions released

when fossil fuels are burned? How should disputes between powerful states and powerful corporations be resolved, and by whom? Each of these questions is, at bottom, a legal question, and the body of law that has developed to answer them is the subject of this book.

International energy and natural resources law is a young discipline in name but an old one in substance. The legal instruments through which states have granted, regulated, taxed, and expropriated mineral and petroleum rights have a history that stretches back more than a century, and the doctrines that frame those transactions — sovereignty, contract, property, state responsibility, and the protection of aliens — are older still. What is new is the recognition that these instruments and doctrines constitute a coherent field of study, with its own sources, principles, institutions, and methods, and with a distinctive set of intellectual problems that cannot be adequately addressed from within any single traditional legal category.

This opening chapter lays the foundation for the remainder of the book. It begins by defining the field and identifying its boundaries. It then traces the historical development of resource governance, explains the internationalisation of energy markets and regulation, and examines the interplay between public law, private law, and transnational legal relations. It considers the relationship between energy law and international economic law, the phenomenon of legal pluralism, and the contemporary challenges confronting global resource governance. The chapter closes by setting out the foundational debates that animate doctoral research and that recur throughout the chapters that follow.

The approach throughout is analytical rather than descriptive. A doctoral candidate in this field must be able to do more than recite the content of treaties and statutes; he or she must be able to explain why the law takes the form that it does, to identify the tensions and contradictions within it, to evaluate competing interpretations, and to propose reasoned reforms. The chapter therefore emphasises the conceptual architecture of the field — the ideas that hold it together and the fault lines along which it is contested.

1.1 Defining Energy and Natural Resources Law

The Problem of Definition

Any attempt to define a field of law must confront a familiar difficulty: legal categories are not natural kinds. They are constructed by scholars, practitioners, courts, and legislators for particular purposes, and their boundaries shift as those purposes change. “Energy law” and “natural resources law” are no exception. Neither phrase appears as a term of art in any multilateral treaty of general application, and neither corresponds to a single body of legislation in most domestic legal systems. Instead, each describes a cluster of rules, principles, institutions, and

practices that share a common subject matter — the ownership, development, transformation, trade, and use of energy sources and natural resources — and that are united by a common set of policy concerns.

A useful working definition, adopted for the purposes of this book, is the following:

> International energy and natural resources law is the body of rules, principles, institutions, and practices — derived from public international law, international economic law, comparative domestic law, and transnational commercial practice — that governs the allocation of rights over energy sources and natural resources, the conduct of activities involving their exploration, development, transformation, transport, trade, and use, and the distribution of the benefits and burdens arising from those activities, in situations that involve more than one legal order.

Several features of this definition deserve comment. First, it is *functional* rather than formal: it defines the field by reference to its subject matter and the problems it addresses, not by reference to a particular source of law. Second, it is *pluralist*: it expressly acknowledges that the relevant rules derive from multiple sources, including domestic legislation and private contracts as well as treaties and custom. Third, it is *distributive*: it identifies the allocation of benefits and burdens as a central concern, reflecting the fact that energy and resource law is fundamentally about who gets what, when, and how. Fourth, it is *transnational*: the field is concerned with situations that involve more than one legal order, whether because a resource straddles a boundary, because an investor is foreign, because a commodity is traded internationally, or because environmental effects cross frontiers.

Energy Law and Natural Resources Law: One Field or Two?

The conjunction "energy and natural resources" requires justification. Energy law, narrowly understood, is concerned with the production, conversion, transmission, distribution, and consumption of energy, whether derived from fossil fuels, nuclear fission, or renewable sources. Natural resources law, narrowly understood, is concerned with the ownership, management, and exploitation of resources found in nature, including minerals, petroleum, water, forests, fisheries, and biological diversity. The two fields overlap substantially — petroleum and natural gas are both natural resources and energy sources — but neither is wholly contained within the other. Renewable electricity generated from wind or solar radiation is an energy source but is not usually described as a natural resource in the extractive sense; copper and lithium are natural resources but are not themselves sources of energy, although they are indispensable to its transmission and storage.

There are strong reasons for treating the two fields together. Historically, the law governing petroleum developed out of the law governing hard-rock minerals, and many of the foundational instruments — concessions, licences, royalties, and the doctrine of permanent sovereignty — are common to both. Functionally, the energy transition is collapsing the distinction: the decarbonisation of energy systems depends on the extraction of critical minerals on an unprecedented scale, on access to land and water for renewable installations, and on the use of geological formations for the storage of carbon dioxide and hydrogen. Institutionally, the same ministries, regulators, and courts often have jurisdiction over both energy and minerals. Doctrinally, the same principles of international law — sovereignty over natural resources, the prohibition of transboundary harm, the protection of foreign investment, and the rights of indigenous peoples — govern both.

This book therefore treats energy law and natural resources law as a single integrated field, while recognising that particular chapters will emphasise one or the other. The integrated approach has the further advantage of exposing the connections between subjects that are too often studied in isolation: between petroleum fiscal regimes and renewable energy incentives, for example, or between mining law and climate law.

Relationship with Adjacent Fields

The field overlaps with several established branches of law, and a doctoral candidate must be able to locate it precisely in relation to each.

Public international law. The general rules of public international law — on sources, treaties, state responsibility, jurisdiction, immunity, and dispute settlement — supply the framework within which international energy and resource law operates. The doctrine of permanent sovereignty over natural resources, discussed in Chapter 2, is a principle of general international law. The law of the sea, international watercourse law, and international environmental law are specialised branches of public international law with direct application to resource activities.

International economic law. The law of international trade, investment, finance, and taxation governs the cross-border movement of energy commodities, capital, and revenues. The World Trade Organization (WTO) agreements apply to trade in energy goods and, to a more limited extent, services; bilateral and multilateral investment treaties protect foreign investors in the energy and mining sectors; and international tax rules shape the fiscal treatment of resource projects. The relationship between energy law and international economic law is examined in section 1.5 below.

International environmental and climate law. Resource activities are among the most significant sources of environmental harm, and international

environmental law imposes obligations of prevention, precaution, assessment, and cooperation that constrain the manner in which they may be carried out. Climate change law, in particular, is reshaping energy law by setting long-term objectives for the reduction of greenhouse gas emissions that are incompatible with the indefinite expansion of fossil fuel production.

Comparative and domestic law. Most of the rules that govern energy and resource activities on a day-to-day basis are domestic: constitutional provisions on resource ownership, petroleum and mining statutes, electricity regulation, environmental legislation, and tax codes. International energy law cannot be understood without close attention to these domestic regimes and to the patterns of convergence and divergence among them, which are explored in Chapter 24.

Transnational commercial law. Energy and resource projects are structured through complex networks of private contracts — joint operating agreements, sales and purchase agreements, engineering and construction contracts, financing agreements — that are frequently governed by English or New York law and subject to international commercial arbitration. The model contracts and standard terms produced by industry associations constitute a form of transnational private ordering that interacts with, and sometimes substitutes for, state regulation.



Figure 1.1 The constituent fields of international energy and natural resources law. The discipline is defined by its subject matter and draws on five overlapping bodies of law. *Source:* prepared for this edition.

The Scope of the Field

The scope of the field, as treated in this book, may be described along three axes.

The first axis is the *resource or energy source* concerned: petroleum (crude oil and natural gas), coal, hard-rock minerals including critical minerals, uranium and nuclear energy, renewable energy sources (solar, wind, hydropower, geothermal, biomass, marine), hydrogen and other energy carriers, electricity, water, and marine resources.

The second axis is the *stage of the value chain*: the allocation of rights to explore and develop resources (upstream), the transport and processing of those resources (midstream), and their sale, distribution, and consumption (downstream), together with the decommissioning of installations and the remediation of affected environments at the end of the project life cycle.

The third axis is the *legal function*: the allocation of property and access rights, the regulation of conduct, the protection of investment, the allocation of commercial risk by contract, the raising and distribution of revenue, the protection of the environment and human rights, and the resolution of disputes.

The chapters of this book are organised principally along the third axis, while particular chapters address specific resources (Chapters 5 to 12) and specific legal functions (Chapters 13 to 27).

Is There a "Lex Petrolea"?

One of the most contested questions in the definition of the field concerns the existence of a distinct transnational legal order governing the international petroleum industry — sometimes described as the *lex petrolea*. The idea has a long pedigree. In the arbitral award in *Kuwait v American Independent Oil Company (Aminoil)*, the tribunal referred to the existence of a body of practice in the oil industry that informed the interpretation of the concession, while stopping short of endorsing an autonomous legal system.[1] Proponents of the *lex petrolea* argue that the repeated use of standard contractual forms, the consistent practice of industry participants, the jurisprudence of arbitral tribunals, and the model contracts of bodies such as the Association of International Energy Negotiators (AIEN) have generated a body of transnational norms that operates partly independently of any national legal system.[2]

Critics respond that the concept is either trivially true — industry practice obviously influences the interpretation of industry contracts — or dangerously overstated, in that it purports to detach contractual relations from the legal systems that ultimately give them force, and that it tends to privilege the interests

[1] *Government of the State of Kuwait v American Independent Oil Company (Aminoil)*, Award of 24 March 1982, (1982) 21 ILM 976.

[2] See R. Doak Bishop, "International Arbitration of Petroleum Disputes: The Development of a *Lex Petrolea*" (1998) 23 *Yearbook of Commercial Arbitration* 1131; and Thomas C.C. Childs, "Update on *Lex Petrolea*: The Continuing Development of Customary Law Relating to International Oil and Gas Exploration and Production" (2011) 4 *Journal of World Energy Law and Business* 214.

of international oil companies over those of host states and affected communities. The debate is significant for doctoral researchers because it raises fundamental questions about the nature of law, the legitimacy of private ordering, and the relationship between state and non-state sources of normativity. It is revisited in Chapter 2 in the context of soft law and industry standards and in Chapter 26 in the context of arbitral practice.

1.2 Historical Development of Resource Governance

A historical perspective is indispensable to an understanding of contemporary energy and resource law. Many of the field's defining doctrines — the stabilisation clause, the doctrine of permanent sovereignty, the standard of compensation for expropriation, the treatment of production sharing contracts — were forged in specific historical controversies, and their meaning cannot be grasped without appreciating the circumstances in which they arose. This section traces five broad phases in the development of resource governance.

Phase One: The Classical Concession Era (c. 1900–1950)

The modern international petroleum industry was built on the classical concession. In 1901, William Knox D'Arcy obtained from the Shah of Persia a concession conferring the exclusive right to explore for, produce, and export petroleum over approximately three-quarters of the territory of Persia for a period of sixty years, in return for a modest initial payment and a share of annual net profits. The D'Arcy concession became the template for a generation of agreements across the Middle East, Latin America, and Asia.

The classical concession had several characteristic features. It covered vast areas, often a substantial proportion of the host state's territory. It endured for very long periods, frequently between sixty and ninety-nine years. It conferred upon the concessionaire not merely the right to extract petroleum but ownership of the petroleum once produced, together with broad managerial control over all aspects of operations. The host state's financial return was limited, typically to a fixed royalty per ton of production, and the state had little or no involvement in decisions concerning the rate of production, the pricing of crude oil, or the allocation of markets. The concessions were frequently negotiated between unequal parties, in circumstances in which the host state lacked the geological information, technical expertise, and bargaining power necessary to secure favourable terms.

The legal character of these instruments was ambiguous. From the perspective of the international oil companies and their home states, a concession was a contract that created vested rights protected by international law, and any unilateral modification or termination by the host state constituted a wrong for which compensation was owed. From the perspective of host states, the concession was an act of sovereign authority — an administrative grant — that the state could

revise in the public interest. This tension between contract and sovereignty is one of the enduring themes of the field.

During this period the industry was dominated by a small number of vertically integrated companies — later known as the "Seven Sisters" — which controlled exploration, production, refining, transportation, and marketing. Arrangements such as the Red Line Agreement of 1928, under which the partners in the Iraq Petroleum Company agreed not to compete with one another for concessions within a defined area of the former Ottoman Empire, and the Achnacarry or "As-Is" Agreement of the same year, which sought to stabilise market shares, illustrate the degree to which the international oil market was organised through private arrangements rather than public regulation.

Phase Two: Nationalisation and the Assertion of Sovereignty (1938–1960s)

The first major challenge to the concession system came from Latin America. In 1938, Mexico nationalised the holdings of foreign oil companies following a prolonged labour dispute and a confrontation with the companies over compliance with a decision of the Mexican Supreme Court. The Mexican nationalisation prompted a celebrated exchange of diplomatic notes in which the United States Secretary of State, Cordell Hull, asserted that international law required "prompt, adequate, and effective" compensation for the expropriation of foreign property — the formula that became known as the Hull standard. Mexico rejected the claim that international law imposed any such requirement, maintaining that compensation was a matter for domestic law. The dispute was eventually settled by agreement, but the doctrinal disagreement it exposed persisted for decades.

The second major challenge came in Iran. In 1951, the Iranian parliament enacted legislation nationalising the assets of the Anglo-Iranian Oil Company. The United Kingdom instituted proceedings before the International Court of Justice, but the Court held that it lacked jurisdiction because the relevant Iranian declaration accepting the Court's compulsory jurisdiction was limited to disputes concerning treaties, and the concession of 1933 was not a treaty between Iran and the United Kingdom.[3] The *Anglo-Iranian Oil Co* case is significant for its holding that a concession agreement between a state and a foreign company is not an international treaty, notwithstanding the involvement of the company's home state in its negotiation.

The same period witnessed the gradual replacement of fixed royalty arrangements with profit-sharing formulas. Beginning with Venezuela in 1943 and spreading to Saudi Arabia in 1950, the so-called "fifty-fifty" principle — under

[3] *Anglo-Iranian Oil Co (United Kingdom v Iran)*, Preliminary Objection, Judgment, ICJ Reports 1952, p. 93.

which the host government received approximately half of the profits from petroleum operations through a combination of royalties and income taxes — became the industry norm. The fifty-fifty principle represented a significant shift in the distribution of resource rents, but it left the structure of the concession — and the managerial control of the companies — largely intact.

Phase Three: Decolonisation and the New International Economic Order (1960–1980)

The third phase was shaped by decolonisation and the emergence of a large number of newly independent states determined to assert control over their natural resources. Three developments were decisive.

First, in September 1960, Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela founded the Organization of the Petroleum Exporting Countries (OPEC). OPEC's initial objective was to resist reductions in posted prices unilaterally imposed by the companies, but it rapidly became a forum through which producing states coordinated their policies on pricing, taxation, and participation.

Second, the General Assembly of the United Nations adopted a series of resolutions articulating the principle of permanent sovereignty over natural resources. The most important of these, Resolution 1803 (XVII) of 14 December 1962, declared that "the right of peoples and nations to permanent sovereignty over their natural wealth and resources must be exercised in the interest of their national development and of the well-being of the people of the State concerned", while also providing that in cases of nationalisation "the owner shall be paid appropriate compensation, in accordance with the rules in force in the State taking such measures in the exercise of its sovereignty and in accordance with international law", and that foreign investment agreements "freely entered into by or between sovereign States shall be observed in good faith".[4] Resolution 1803 represented a carefully negotiated compromise between capital-exporting and capital-importing states, and it is widely regarded as reflecting customary international law in significant respects.

Third, in the early 1970s, producing states moved from profit-sharing to participation and outright nationalisation. Libya nationalised the interests of several foreign companies between 1971 and 1974; Algeria took a majority interest in French oil companies in 1971; Iraq nationalised the Iraq Petroleum Company in 1972; and the states of the Persian Gulf negotiated progressive participation agreements that culminated in full state ownership of their producing industries. The oil price increases of 1973–1974, associated with the Arab oil embargo and

[4] UN General Assembly Resolution 1803 (XVII), "Permanent Sovereignty over Natural Resources", 14 December 1962, paras 1, 4, and 8.

OPEC's assertion of control over pricing, transformed the economics of the industry and the balance of power between producing states and consuming states.

In 1974, the General Assembly adopted the Declaration on the Establishment of a New International Economic Order and the Charter of Economic Rights and Duties of States.[5] Article 2 of the Charter asserted the right of every state "to nationalize, expropriate or transfer ownership of foreign property, in which case appropriate compensation should be paid by the State adopting such measures, taking into account its relevant laws and regulations and all circumstances that the State considers pertinent", and provided that any controversy over compensation "shall be settled under the domestic law of the nationalizing State and by its tribunals", unless otherwise agreed. Unlike Resolution 1803, the Charter omitted any reference to international law as the governing standard for compensation and was adopted over the opposition or abstention of most capital-exporting states. Its status as a statement of customary international law has accordingly been doubted.

The legal consequences of the nationalisations of this period were addressed in a series of landmark arbitral awards, often referred to collectively as the "Libyan oil arbitrations" — *BP v Libya* (1973), *Texaco v Libya* (1977), and *LIAMCO v Libya* (1977) — together with the *Aminoil* award of 1982. These awards, which are examined in detail in Chapters 3 and 14, grappled with the legal status of concession contracts, the effect of stabilisation clauses, the applicable law, and the appropriate standard of compensation. Their reasoning remains foundational to the modern law of foreign investment.

[5] UN General Assembly Resolution 3201 (S-VI), "Declaration on the Establishment of a New International Economic Order", 1 May 1974; UN General Assembly Resolution 3281 (XXIX), "Charter of Economic Rights and Duties of States", 12 December 1974.

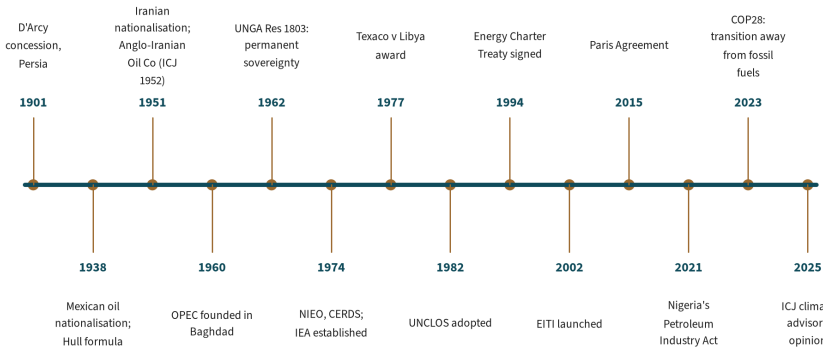


Figure 1.2 Selected milestones in the development of international energy and natural resources law, 1901–2025. *Source:* prepared for this edition.

Phase Four: Liberalisation, Investment Protection, and Market Integration (1980–2005)

The fourth phase was characterised by a reversal of the statist tendencies of the preceding period. The collapse of oil prices in the mid-1980s, the debt crisis in the developing world, the end of the Cold War, and the ascendancy of market-oriented economic policies led many states to reopen their resource sectors to foreign investment. Production sharing contracts — first used in Indonesia in the 1960s — became the dominant contractual form in many developing producer states, allowing governments to retain formal ownership of petroleum while engaging foreign companies as contractors entitled to a share of production. Mining codes across Africa and Latin America were rewritten, often with the support of international financial institutions, to attract foreign capital through security of tenure, fiscal stability, and liberal repatriation of profits.

At the same time, a dense network of bilateral investment treaties (BITs) emerged. From a few hundred in the 1980s, the number of BITs grew to more than two thousand by the early 2000s. These treaties conferred upon foreign investors substantive protections — fair and equitable treatment, full protection and security, protection against expropriation without compensation, national and most-favoured-nation treatment — and, crucially, the right to bring claims directly against host states before international arbitral tribunals, typically under the auspices of the International Centre for Settlement of Investment Disputes (ICSID) established by the Washington Convention of 1965. The extension

of investor–state arbitration transformed the enforcement of investor rights by removing the need for diplomatic protection by the investor's home state.

In the energy sector specifically, the Energy Charter Treaty (ECT), signed in Lisbon in December 1994 and in force since April 1998, established a multilateral framework for investment protection, trade, transit, and energy efficiency among the states of Europe and the former Soviet Union, together with a number of states in Asia. The ECT became the most frequently invoked investment treaty in the world, a fact that would later make it a central battleground in debates over the compatibility of investment protection with climate policy.

The liberalisation of domestic energy markets proceeded in parallel. In the electricity and gas sectors, the United Kingdom privatised its state-owned utilities in the late 1980s and 1990s; the United States introduced open access to transmission networks through Federal Energy Regulatory Commission (FERC) Order No. 888 of 1996; and the European Union adopted successive legislative packages, beginning with the first Electricity Directive of 1996, designed to create an internal energy market through the unbundling of vertically integrated utilities and the introduction of competition in generation and supply.

Phase Five: Resource Nationalism, Sustainability, and the Energy Transition (2005–present)

The fifth phase, which continues today, is marked by a complex and sometimes contradictory set of trends.

First, the commodity price boom of the 2000s prompted a revival of resource nationalism. Venezuela, Bolivia, Ecuador, and Russia, among others, increased state participation, renegotiated contracts, or expropriated foreign interests, generating a new wave of investment arbitrations, several of which resulted in awards of billions of United States dollars. Resource nationalism also took subtler forms, including increased royalties and windfall taxes, local content requirements, export restrictions on unprocessed minerals, and mandatory state participation in new projects.

Second, the international community increasingly emphasised transparency, good governance, and the equitable distribution of resource revenues. The Extractive Industries Transparency Initiative (EITI), launched in 2002 and now implemented in more than fifty countries, requires the disclosure of payments by companies to governments and of revenues received by governments from the extractive sector. Supply chain due diligence standards for conflict minerals, notably the OECD *Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas*, sought to address the link between mineral extraction and armed conflict. The United Nations *Guiding Principles on Business and Human Rights*, endorsed by the Human Rights Council in

2011, articulated the responsibility of business enterprises, including extractive companies, to respect human rights.

Third, and most consequentially, climate change has moved from the margins to the centre of energy law. The Paris Agreement of 2015 set the objective of holding the increase in global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the increase to 1.5°C. At the twenty-eighth Conference of the Parties to the United Nations Framework Convention on Climate Change (COP28) in Dubai in 2023, the parties for the first time called in the outcome of the first global stocktake for "transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner". Climate litigation has proliferated in domestic courts, regional human rights bodies, and international tribunals, culminating in advisory opinions from the International Tribunal for the Law of the Sea (2024), the Inter-American Court of Human Rights (2025), and the International Court of Justice (2025) on the obligations of states in respect of climate change. These developments are examined in Chapter 18.

Fourth, the energy transition has created new legal frontiers: the regulation of renewable energy, hydrogen, energy storage, carbon capture and storage, and offshore wind; the governance of critical minerals essential to batteries, electric vehicles, and grid infrastructure; and the management of stranded assets and the decommissioning of fossil fuel infrastructure. Simultaneously, geopolitical events — notably the Russian invasion of Ukraine in February 2022 and the resulting disruption of European gas supply — have restored energy security to the forefront of policy and have demonstrated that the transition will be shaped as much by security considerations as by environmental ones.

KEY DOCTRINAL POINT

The history of resource governance is not a linear progression from one paradigm to another. Elements of each phase coexist in contemporary law. Classical concession-style licences remain in use in many jurisdictions; the doctrine of permanent sovereignty retains its normative force; investment treaties continue to protect investors even as they are reformed or terminated; and the imperatives of the energy transition are layered on top of these older structures rather than replacing them. A doctoral analysis must therefore be historically literate: it must recognise which layer of the law a given rule belongs to and how that rule has been reinterpreted over time.

1.3 Internationalisation of Energy Markets and Resource Regulation

The Global Character of Energy Markets

Energy markets are among the most internationalised of all economic sectors. Crude oil is a global commodity, traded on international exchanges and priced by reference to benchmark grades such as Brent and West Texas Intermediate. A substantial proportion of the world's oil production crosses at least one international border before it is consumed, often passing through maritime chokepoints such as the Strait of Hormuz and the Strait of Malacca. Natural gas markets, historically regional because of the constraints of pipeline transport, have become increasingly integrated through the growth of the liquefied natural gas (LNG) trade, which permits gas to be shipped across oceans and redirected in response to price signals. Coal is traded internationally in large volumes. Electricity, while still predominantly national, is increasingly traded across borders through interconnectors and regional power pools. The supply chains for renewable energy technologies — solar panels, wind turbines, batteries — are global, with manufacturing concentrated in a small number of states.

The internationalisation of markets has legal consequences. Domestic regulatory decisions in one state — on export restrictions, subsidies, environmental standards, or sanctions — can have profound effects on prices and supplies in others. Commercial disputes arising from cross-border transactions require rules on applicable law, jurisdiction, and the enforcement of judgments and awards. Infrastructure that crosses borders, such as pipelines and interconnectors, requires intergovernmental agreements to address transit, tariffs, and the allocation of regulatory authority.

Mechanisms of Internationalisation

The internationalisation of energy and resource regulation has proceeded through several mechanisms.

Treaty-making. States have concluded treaties addressing particular aspects of energy and resource activities: investment protection (BITs and the ECT), trade (the WTO agreements and regional trade agreements), transboundary pipelines (for example, the intergovernmental agreements underlying the Baku–Tbilisi–Ceyhan pipeline and the West African Gas Pipeline), nuclear safety and liability, the law of the sea, and environmental protection.

Institutional cooperation. International organisations such as the International Energy Agency (IEA), the International Atomic Energy Agency (IAEA), the International Renewable Energy Agency (IRENA), OPEC, and the Gas Exporting Countries Forum provide forums for the coordination of national policies, the exchange of information, and, in some cases, the adoption of binding

standards. Regional bodies — the Energy Community in South-East Europe, the West African Power Pool, the Southern African Power Pool — promote regional market integration.

Regulatory convergence. States often borrow regulatory models from one another, whether through voluntary emulation, the conditions attached to development finance, or the technical assistance of international organisations. The spread of independent regulatory agencies, of unbundling requirements in electricity, and of model petroleum agreements reflects this process of regulatory transplantation, which is examined in Chapter 24.

Private ordering. The standard contracts produced by industry associations — the AIEN Model Joint Operating Agreement, the International Federation of Consulting Engineers (FIDIC) suites of construction contracts, and the master agreements used in energy trading — create transnational norms that operate across jurisdictions.

Extraterritorial regulation. Some states regulate conduct outside their territory through legislation with extraterritorial effect. Examples include anti-corruption statutes such as the United States Foreign Corrupt Practices Act of 1977 and the United Kingdom Bribery Act 2010, sanctions regimes targeting the energy sectors of particular states, and supply chain due diligence obligations such as those contained in the European Union's Corporate Sustainability Due Diligence Directive and Batteries Regulation. These are discussed in Chapters 2 and 22.

The Limits of Internationalisation

Notwithstanding these developments, energy and resource regulation remains in important respects national. There is no comprehensive multilateral energy treaty of universal application; the ECT, the most ambitious attempt to create such an instrument, never attracted the participation of major producers and consumers such as the United States, China, Saudi Arabia, or Brazil, and its membership has contracted in recent years following withdrawals by the European Union and several of its member states. Petroleum, mining, and electricity remain subject primarily to domestic legislation, and the doctrine of permanent sovereignty ensures that the fundamental decisions about whether, when, and on what terms to develop natural resources are reserved to the state in which they are located. The internationalisation of energy law is thus partial, uneven, and contested — characteristics that make it a fertile subject for doctoral research.

1.4 Public Law, Private Law, and Transnational Legal Relations

The Public–Private Divide in Energy and Resource Law

One of the distinctive features of energy and resource law is the extent to which it straddles the conventional divide between public and private law. On the public side, the state acts as sovereign: it owns or controls subsurface resources, grants licences, sets fiscal terms, regulates conduct, protects the environment, and resolves disputes through its courts. On the private side, the state frequently acts as a commercial party: it enters into contracts with investors, participates in joint ventures through national oil or mining companies, sells resources on international markets, and borrows money to finance its share of project costs.

The dual character of the state — as sovereign regulator and as contracting party — generates persistent legal difficulties. When a state that has entered into a contract with a foreign investor subsequently changes its tax laws or environmental regulations in a manner that adversely affects the investor, is it acting as a sovereign exercising its regulatory powers, or as a contracting party in breach of its contractual obligations? The answer may determine whether the investor has a remedy, which tribunal has jurisdiction, and what law applies. The distinction between acts *jure imperii* and acts *jure gestionis*, familiar from the law of state immunity, and the distinction in investment law between a mere breach of contract and a breach of treaty, are both attempts to grapple with this problem.

The State Contract

The "state contract" — a contract between a state (or a state entity) and a foreign private party — is the central legal instrument of international resource development. Petroleum concessions, production sharing contracts, mining agreements, host government agreements for pipelines, and power purchase agreements with state-owned utilities are all species of state contract.

The legal status of state contracts has been debated for more than a century. Three principal positions may be distinguished.

The first position holds that state contracts are governed exclusively by the domestic law of the host state. On this view, the state retains its sovereign power to modify the contract by legislation, and the investor's protection lies only in such guarantees as domestic law provides. This position was associated with the Calvo doctrine in Latin America and with the claims of the New International Economic Order.

The second position holds that state contracts may be "internationalised" — that is, subjected to international law or to general principles of law — by the choice of the parties, by the inclusion of stabilisation and arbitration clauses, or by the nature of the contract as an economic development agreement. This

position found its fullest expression in the award of Sole Arbitrator René-Jean Dupuy in *Texaco v Libya*, which held that the concession contracts at issue had been internationalised and that Libya's nationalisation measures were a breach of its contractual obligations under international law.[6]

The third position, now widely accepted, holds that the contract is governed by the law chosen by the parties (often the host state's law, sometimes supplemented by international law or general principles), but that the investor may also enjoy independent protection under international law — most importantly through investment treaties — which may be invoked even where the contract itself is governed by domestic law. The distinction between contract claims and treaty claims, articulated by the *ad hoc* Committee in *Compañía de Aguas del Aconquija SA and Vivendi Universal v Argentina* (*Vivendi I*), is now a cornerstone of investment arbitration: "whether there has been a breach of the BIT and whether there has been a breach of contract are different questions".[7]

Transnational Legal Relations

Beyond the public–private divide lies a further category of relations that is neither purely public nor purely private, neither purely national nor purely international. Philip Jessup, writing in 1956, coined the term "transnational law" to include "all law which regulates actions or events that transcend national frontiers", embracing both public and private international law as well as "other rules which do not wholly fit into such standard categories".[8] Energy and resource activities exemplify transnational legal relations. A single offshore gas project may involve a host government agreement with the state, a production sharing contract with the national oil company, a joint operating agreement among the co-venturers, engineering contracts with international contractors, LNG sales agreements with buyers in several countries, financing agreements with commercial banks and export credit agencies, and an investment treaty between the host state and the home states of the investors. Each relationship is governed by a different combination of national and international rules, and disputes may be resolved in different forums.

The transnational character of these relations has three important implications. First, it means that no single legal order governs the project as a whole; the applicable law must be determined relationship by relationship. Second, it creates the risk of conflicting obligations, as when an environmental obligation imposed by domestic law conflicts with a stabilisation clause in a contract, or a

[6] *Texaco Overseas Petroleum Company and California Asiatic Oil Company v Government of the Libyan Arab Republic*, Award on the Merits of 19 January 1977, (1978) 17 ILM 1; (1979) 53 ILR 389.

[7] *Compañía de Aguas del Aconquija SA and Vivendi Universal v Argentine Republic*, ICSID Case No ARB/97/3, Decision on Annulment, 3 July 2002, para 96.

[8] Philip C. Jessup, *Transnational Law* (Yale University Press 1956), p. 2.

sanctions regime imposed by one state conflicts with a contractual obligation to supply. Third, it confers significant influence upon private actors — companies, banks, industry associations, arbitrators — in the creation and interpretation of norms, raising questions of legitimacy and accountability that recur throughout this book.

1.5 The Relationship Between Energy Law and International Economic Law

Energy as an Exception in Trade Law

The relationship between energy law and international economic law is close but historically uneasy. When the General Agreement on Tariffs and Trade (GATT) was negotiated in 1947, energy products were not excluded from its scope, but in practice trade in oil and gas was largely left outside the multilateral trading system. Many major producers were not parties to the GATT; the trade in crude oil was conducted largely within vertically integrated companies; and energy policy was regarded as so closely bound up with national security that states were unwilling to subject it to multilateral discipline. The GATT security exception in Article XXI, which permits measures "relating to fissionable materials or the materials from which they are derived" and measures "taken in time of war or other emergency in international relations", reinforced this sense of energy as a domain of sovereign prerogative.

The establishment of the WTO in 1995 and the accession of major energy producers — including Saudi Arabia in 2005 and the Russian Federation in 2012 — brought energy trade more firmly within the multilateral framework. WTO disputes have since addressed export restrictions on raw materials (*China — Raw Materials* and *China — Rare Earths*), local content requirements in renewable energy programmes (*Canada — Renewable Energy/Feed-in Tariff Program* and *India — Solar Cells*), and the European Union's energy market regulation (*EU — Energy Package*). These disputes are examined in Chapter 13.

Investment Law as the Primary Interface

If trade law has historically played a limited role in energy, investment law has played a central one. Energy and mining projects are capital-intensive, long-lived, and dependent on the stability of the host state's legal and fiscal regime; they are therefore particularly vulnerable to what economists call the "obsolescing bargain" — the tendency for the host state's bargaining power to increase once the investor's capital has been sunk. Investment treaties and investor–state arbitration were designed in large part to mitigate this risk. Statistics compiled by ICSID consistently show that the oil, gas, and mining sector, together with electric power

and other energy, account for a larger share of registered cases than any other economic sector.

The obsolescing bargain theory, associated with the work of Raymond Vernon, explains much of the dynamic of resource governance. At the time of the initial investment decision, the host state competes with other states for scarce capital and expertise and must offer attractive terms. Once the investment has been made, however, the investor's assets are immobile and the host state may seek to renegotiate the terms in its favour — through increased taxation, contract renegotiation, or expropriation. Investors anticipate this risk and demand protections: stabilisation clauses, international arbitration, investment treaty coverage, and political risk insurance. The law of international energy investment can be understood, in significant part, as an attempt to manage the obsolescing bargain.

Convergence and Tension

The relationship between energy law and international economic law is characterised by both convergence and tension. Convergence occurs where economic law supports energy policy objectives — for example, where investment protection encourages the capital flows necessary for the energy transition. Tension occurs where economic law constrains the regulatory autonomy that states need to pursue energy and climate objectives — for example, where investors challenge the phase-out of coal-fired power or the withdrawal of renewable energy subsidies, or where trade rules are invoked against local content requirements designed to build domestic manufacturing capacity. The phase-out of coal-fired power in the Netherlands gave rise to ECT claims by RWE and Uniper (the latter subsequently withdrawn), and the retroactive reduction of renewable energy incentives in Spain generated dozens of ECT arbitrations. The resulting debates about regulatory chill, the right to regulate, and the reform of investment treaties are examined in Chapter 14.

1.6 Legal Pluralism and Competing Regulatory Authorities

The Concept of Legal Pluralism

Legal pluralism describes situations in which two or more legal orders coexist in the same social field. Originally developed by legal anthropologists to describe the coexistence of state law and customary law in colonial and post-colonial societies, the concept has been extended to describe the coexistence of national, international, regional, and private normative orders in a globalised world. Energy and resource governance is a paradigmatic site of legal pluralism.

At least five layers of normative authority may be identified in a typical resource project:

1. **Customary and community norms.** In many resource-rich regions, land and resources are held under customary tenure systems that predate, and

coexist with, state law. In Nigeria, for example, customary land tenure was overlaid by the Land Use Act 1978, which vested land in each state in the Governor, while section 44(3) of the Constitution of the Federal Republic of Nigeria 1999 vests the entire property in and control of all minerals, mineral oils, and natural gas in the Government of the Federation. Communities in the Niger Delta nevertheless continue to assert claims to land and resources grounded in custom, and these claims shape the conduct of operations and the content of community development agreements.

2. **Subnational law.** In federal states, constitutional authority over resources may be divided between the federal government and the constituent units. In Canada, the provinces own and regulate most natural resources under section 92A of the Constitution Act 1867 (inserted in 1982); in Australia, the states own onshore minerals and petroleum; in the United States, mineral rights may be held privately, by the states, or by the federal government. In Nigeria, the Electricity Act 2023, enacted following a constitutional amendment, empowers states to regulate electricity markets within their territories.
3. **National law.** Constitutional provisions, petroleum and mining statutes, environmental legislation, tax law, and administrative law form the core of the regulatory framework.
4. **International and regional law.** Treaties, customary international law, and the law of regional organisations (such as the European Union, the Economic Community of West African States, and the African Union) impose obligations on states and, in some cases, confer rights on private parties.
5. **Transnational private regulation.** Industry standards, model contracts, voluntary codes of conduct, certification schemes, and the environmental and social performance standards of lenders — notably the International Finance Corporation (IFC) *Performance Standards on Environmental and Social Sustainability* and the Equator Principles adopted by commercial banks — impose requirements on companies that may exceed those of national law.

Sources of Conflict

The coexistence of multiple normative orders generates conflicts of several kinds. *Vertical* conflicts arise between different levels of authority — for example, between federal and state governments over the ownership of offshore resources, as in Nigeria's resource-control litigation, in which the Supreme Court held in *Attorney-General of the Federation v Attorney-General of Abia State* that for the purposes of revenue allocation the seaward boundary of a littoral state was the

low-water mark of its land surface.[9] *Horizontal* conflicts arise between different regimes at the same level — for example, between investment treaty obligations and environmental obligations. *Public–private* conflicts arise between state regulation and private standards or contracts. *Jurisdictional* conflicts arise between courts and tribunals asserting authority over the same dispute, as when an investor pursues parallel proceedings before domestic courts, commercial arbitration, and investment treaty arbitration.

Managing Pluralism

Legal systems have developed a range of techniques for managing these conflicts: hierarchical rules of priority (constitutional supremacy, the primacy of European Union law), principles of interpretation (systemic integration under Article 31(3)(c) of the Vienna Convention on the Law of Treaties, the presumption against conflict), doctrines of comity and deference (the margin of appreciation, the police powers doctrine in investment law), procedural mechanisms (fork-in-the-road and waiver clauses, *lis pendens*, *res judicata*), and institutional coordination (memoranda of understanding between regulators, joint commissions).

The International Law Commission's study of the fragmentation of international law, completed in 2006 under the chairmanship of Martti Koskenniemi, remains the most authoritative analysis of these techniques at the international level.[10] For energy and resource law, the key insight of the fragmentation debate is that specialised regimes — investment law, trade law, environmental law, human rights law — do not operate in isolation; they remain embedded in general international law, and their interpretation must take account of other relevant rules applicable between the parties.

> **COMPARATIVE PERSPECTIVE**
>
> **Federalism and resource authority in three jurisdictions.** In *Canada*, section 92A of the Constitution Act 1867 confirms provincial legislative authority over the exploration, development, conservation, and management of non-renewable natural resources, subject to federal paramountcy for interprovincial export. In *Nigeria*, section 44(3) of the 1999 Constitution vests ownership and control of minerals and petroleum in the Federal Government, and item 39 of the Exclusive Legislative List reserves "mines and minerals, including oil fields, oil mining, geological surveys and natural gas" to the National Assembly; the derivation principle in section 162(2) allocates not less than thirteen per cent of revenue accruing from natural resources

[9] *Attorney-General of the Federation v Attorney-General of Abia State & 35 Others* (2002) 6 NWLR (Pt 764) 542 (Supreme Court of Nigeria).

[10] International Law Commission, *Fragmentation of International Law: Difficulties Arising from the Diversification and Expansion of International Law*, Report of the Study Group finalised by Martti Koskenniemi, UN Doc A/CN.4/L.682 (13 April 2006).

to the states of origin. In the *United States*, subsurface minerals may be privately owned and severed from the surface estate, with federal and state governments owning substantial mineral estates in their own right and regulating activities on federal and state lands respectively. These differences shape the bargaining power of communities, the structure of fiscal regimes, and the incidence of resource conflicts.

1.7 Contemporary Challenges in Global Resource Governance

The field confronts a set of challenges that are without precedent in their scale and complexity. Six are of particular importance.

The Energy Transition and Stranded Assets

The first challenge is the transition from an energy system dominated by fossil fuels to one based predominantly on low-carbon sources. The legal implications are profound. Petroleum and coal licences granted for decades may become economically or legally unviable before their expiry, raising questions about compensation, decommissioning liability, and the protection of legitimate expectations. Investors in fossil fuel infrastructure may seek to rely on investment treaties to challenge phase-out measures. Producer states whose economies depend on hydrocarbon revenues face fiscal crises and must manage the social consequences of transition. The concept of "stranded assets" — assets that suffer from premature write-downs or devaluation as a result of the transition — has migrated from financial analysis into law, influencing disclosure obligations, directors' duties, and the interpretation of investment protections.

Critical Minerals and the New Extractivism

The second challenge arises from the paradox that decarbonisation requires more extraction, not less. The IEA has repeatedly projected that demand for minerals such as lithium, cobalt, nickel, copper, graphite, and rare earth elements will grow severalfold under scenarios consistent with the Paris Agreement. These minerals are geographically concentrated, and their processing is concentrated still further. States have responded with industrial policies, export restrictions, strategic stockpiles, and bilateral critical mineral agreements. Communities in mining regions confront the risk that the green transition will reproduce the environmental degradation and human rights abuses associated with earlier extractive booms. Chapter 8 addresses these issues in detail.

Energy Access and Energy Poverty

The third challenge is that a substantial proportion of humanity still lacks access to modern energy services. Hundreds of millions of people, the majority in sub-Saharan Africa, lack access to electricity, and far more lack access to clean cooking fuels. Sustainable Development Goal 7 commits states to ensure access

to affordable, reliable, sustainable, and modern energy for all by 2030. Achieving universal access while simultaneously decarbonising energy systems requires legal and regulatory frameworks that mobilise investment in decentralised and grid-based solutions, protect consumers, and allocate the costs of transition fairly. The concept of energy justice, explored in Chapter 23, provides a framework for analysing these distributional questions.

Geopolitics and Energy Security

The fourth challenge is the re-emergence of energy as an instrument of geopolitical competition. The disruption of gas supplies to Europe following the invasion of Ukraine, the use of sanctions against major energy exporters, attacks on energy infrastructure, and competition over critical mineral supply chains have demonstrated that energy security remains a central concern of state policy. The law governing sanctions, export controls, emergency powers, strategic reserves, and the protection of critical infrastructure — examined in Chapter 25 — has accordingly grown in importance.

Legitimacy of Investor–State Dispute Settlement

The fifth challenge concerns the legitimacy of the system of investor–state dispute settlement (ISDS). Critics argue that ISDS privileges foreign investors over domestic stakeholders, constrains the regulatory autonomy of states, produces inconsistent decisions, and lacks transparency and accountability. Reform efforts proceed in several forums: the United Nations Commission on International Trade Law (UNCITRAL) Working Group III, the European Union's proposal for a multilateral investment court, the modernisation of the ECT, and the renegotiation or termination of BITs by individual states. The outcome of these reforms will shape the legal environment for energy investment for decades.

Technological Change and Digitalisation

The sixth challenge arises from technological change. Hydrogen, advanced nuclear reactors, carbon capture, battery storage, offshore floating wind, and deep-sea mining all raise novel regulatory questions. Digitalisation — smart meters, distributed energy resources, automated trading, artificial intelligence in grid management — raises questions of data ownership, cybersecurity, and liability. Regulatory frameworks designed for centralised, fossil-fuel-based systems must be adapted to decentralised, digital, and decarbonised systems. Chapters 12 and 30 address these developments.

1.8 Doctoral Research Perspectives and Foundational Debates

The Nature of Doctoral Contribution

Doctoral research is distinguished from other forms of legal writing by the requirement of an original contribution to knowledge. In international energy

and natural resources law, such a contribution may take several forms: the identification and analysis of a new legal problem; a new interpretation of existing rules or cases; the application of a theoretical framework to a body of law; a comparative study revealing previously unrecognised patterns; an empirical investigation of how the law operates in practice; or a reasoned proposal for reform. Chapter 28 discusses research methodology in detail. Here, the purpose is to identify the foundational debates within which most doctoral research in the field is situated.

Debate One: Sovereignty versus Stability

The most enduring debate in the field concerns the tension between the sovereign right of states to regulate their natural resources in the public interest and the need of investors for legal and fiscal stability. The debate manifests itself in disputes over stabilisation clauses, the standard of compensation for expropriation, the scope of the fair and equitable treatment standard, and the legitimacy of contract renegotiation. Is the doctrine of permanent sovereignty compatible with binding long-term commitments to foreign investors? Can a state contract away its legislative power? How should the law balance the legitimate expectations of investors against the evolving needs of the host society?

Debate Two: Contract versus Regulation

A second debate concerns the relative roles of contract and regulation in governing resource activities. In some jurisdictions, the terms of resource development are set primarily by individually negotiated contracts; in others, primarily by general legislation and regulation applicable to all licensees. Each approach has advantages and drawbacks. Contracts offer flexibility and can be tailored to the circumstances of particular projects, but they are vulnerable to corruption, asymmetries of information, and lack of transparency. Regulation offers predictability, equality of treatment, and democratic accountability, but may be less responsive to project-specific risks. The trend in many jurisdictions, including Nigeria under the Petroleum Industry Act 2021, has been towards greater reliance on legislative frameworks, but contracts remain central.

Debate Three: Efficiency versus Equity

A third debate concerns the objectives of resource law. Economic approaches emphasise efficiency — the maximisation of resource rents, the minimisation of transaction costs, and the allocation of risk to the party best able to bear it. Equity-oriented approaches emphasise the fair distribution of benefits and burdens among states, investors, communities, and future generations, and the protection of the rights of affected persons. The "resource curse" literature, which examines why resource-rich states often experience slower growth, weaker institutions, and greater conflict than resource-poor states, has highlighted the importance of

institutional quality and has prompted reforms focused on transparency, revenue management, and accountability.

Debate Four: Development versus Sustainability

A fourth debate concerns the relationship between resource development and environmental sustainability. Developing states argue that they have a right to exploit their resources to finance development, as industrialised states did before them, and that the burden of climate mitigation should fall primarily on those states responsible for historical emissions. This argument is reflected in the principle of common but differentiated responsibilities and respective capabilities, articulated in the United Nations Framework Convention on Climate Change. At the same time, the physical limits of the carbon budget mean that a substantial proportion of known fossil fuel reserves cannot be burned if the temperature goals of the Paris Agreement are to be met. How should the right to development be reconciled with the imperative of decarbonisation? Who should bear the costs of leaving resources in the ground?

Debate Five: State-Centrism versus Transnational Governance

A fifth debate concerns the appropriate unit of analysis. Traditional approaches treat the state as the principal actor and source of law. Transnational approaches emphasise the role of corporations, international organisations, civil society, communities, and private standard-setters in creating and enforcing norms. The rise of corporate climate litigation, supply chain due diligence obligations, and private sustainability standards suggests that governance is increasingly polycentric. Doctoral research must decide how to account for this complexity without losing analytical clarity.

LEADING AUTHORITY

Texaco Overseas Petroleum Co v Libyan Arab Republic (1977)

Facts. Libya had granted deeds of concession to two American companies under its Petroleum Law of 1955. The concessions contained a stabilisation clause providing that the contractual rights expressly created by the concession should not be altered except by mutual consent, and an arbitration clause. In 1973 and 1974 Libya nationalised the companies' interests. The companies invoked arbitration; Libya refused to participate, and a sole arbitrator was appointed by the President of the International Court of Justice.

Holding. Sole Arbitrator Dupuy held that the concessions were internationalised contracts; that they were governed by principles of Libyan law common to international law and, in the absence of such common principles, by general principles of law; that the stabilisation clause was valid and binding; and that Libya had breached its obligations. Notably, he ordered *restitutio in integrum* — specific performance — as the primary remedy, although the parties subsequently settled.

Significance. The award is the high-water mark of the internationalisation theory. Its analysis of Resolution 1803 as reflecting customary law, and its treatment of the later Charter of Economic Rights and Duties of States as lacking that status, remain influential. Its remedy of restitution has rarely been followed; later tribunals, beginning with *LIAMCO* and *Aminoil*, preferred damages.

PRACTICAL EXERCISE — HYPOTHETICAL SCENARIO

The Kestrel Basin concession (hypothetical). The Republic of Aranda, a fictional coastal state, granted a thirty-year petroleum concession in 1998 to Northwind Energy plc, a company incorporated in a fictional state with which Aranda has a bilateral investment treaty. The concession contains (i) a choice of Arandan law "supplemented by such principles of international law as may be applicable"; (ii) a "freezing" stabilisation clause; and (iii) ICC arbitration seated in Geneva. In 2025, following discoveries of significant gas reserves and a change of government, Aranda enacts a Petroleum Revenue Act that raises the royalty rate from 8 to 15 per cent, imposes a windfall profits tax, and requires all existing concessionaires to convert to production sharing contracts within two years.

Prepare a memorandum of not more than 2,500 words that: (a) identifies each legal order potentially applicable to the dispute; (b) explains the competing characterisations of the measures as sovereign regulation or contractual breach; (c) assesses the legal effect of the stabilisation clause in the light of the doctrine of permanent sovereignty; and (d) recommends a dispute resolution strategy for Northwind, identifying the risks of parallel proceedings. *All parties, places, and instruments in this exercise are fictitious.*

CHAPTER SUMMARY

- International energy and natural resources law is a functional, pluralist, distributive, and transnational field concerned with the allocation of rights over energy sources and natural resources, the regulation of activities involving them, and the distribution of the resulting benefits and burdens.
- The field integrates energy law and natural resources law, a combination justified by shared history, doctrine, institutions, and — increasingly — the material interdependence created by the energy transition.
- Its history may be organised into five overlapping phases: the classical concession era, nationalisation, decolonisation and the New International Economic Order, liberalisation and investment protection, and the current era of resource nationalism, sustainability, and transition.
- Energy markets are highly internationalised, but energy regulation remains predominantly national; internationalisation proceeds through treaties, institutions, regulatory convergence, private ordering, and extraterritorial regulation.

- The dual character of the state as sovereign and contracting party is the source of many of the field's enduring problems, notably the status of state contracts and the distinction between contract and treaty claims.
- Legal pluralism is a defining feature of resource governance, requiring techniques of conflict management drawn from constitutional, international, and private law.
- Doctoral research is situated within five foundational debates: sovereignty versus stability, contract versus regulation, efficiency versus equity, development versus sustainability, and state-centrism versus transnational governance.

DOCTORAL RESEARCH QUESTIONS

1. To what extent does the concept of *lex petrolea* describe a genuine transnational legal order, and what are the implications of its existence or non-existence for the legitimacy of investor–state arbitration?
2. Has the energy transition rendered the integrated study of energy and natural resources law analytically indispensable, or does it obscure important differences between extractive and non-extractive energy sources?
3. How should the obsolescing bargain theory be revised to account for renewable energy projects, in which the host state's bargaining power may evolve differently from that in petroleum projects?
4. Can the techniques developed to manage the fragmentation of international law adequately address conflicts between investment protection and climate obligations, or is structural reform required?
5. What role should customary land tenure and community norms play in the allocation of resource rights in states, such as Nigeria, where the Constitution vests mineral ownership in the federal government?

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Placeholder.